

Heat Transfer Explained

1. Introduction

Heat transfer is the study of how thermal energy moves from one body or region to another because of a temperature difference. It is one of the most important subjects in mechanical engineering because it underpins the design of boilers, heat exchangers, refrigerators, condensers, radiators, engines, turbines, furnaces, and thermal protection systems.

Heat transfer is different from thermodynamics. Thermodynamics tells us how much heat is exchanged between equilibrium states, while heat transfer tells us the rate, mechanism, and spatial distribution of that energy flow.

2. Meaning of heat transfer

Heat transfer is the movement of energy driven by temperature difference. The greater the temperature difference, the greater the tendency for heat to flow from the hotter region to the colder region.

In engineering analysis, the question is not only how much heat is transferred, but also how fast it is transferred, through what mechanism, and under what boundary conditions.

3. Modes of heat transfer

There are three fundamental modes of heat transfer:

- Conduction.
- Convection.
- Radiation.

These three modes often occur simultaneously in real systems, such as heat flow through a wall exposed to air and sunlight.

4. Conduction

Conduction is heat transfer through a material or between bodies in direct contact without bulk movement of the material. It occurs because of molecular interactions and energy exchange between neighbouring particles.

In solids, conduction is especially important because particles are closely packed. Metals conduct heat very well because they have free electrons that transport energy efficiently.

4.1 Fourier's law

For one-dimensional conduction:

$$q_x = -kA \frac{dT}{dx}$$

where q_x is heat transfer rate, k is thermal conductivity, A is area, and $\frac{dT}{dx}$ is the temperature gradient.

4.2 Thermal conductivity

Thermal conductivity measures a material's ability to conduct heat. High k materials such as copper and aluminium transfer heat quickly, while low k materials such as wood, plastic, and insulation transfer heat slowly.

4.3 Types of conduction

- Steady-state conduction.
- Transient conduction.
- One-dimensional, two-dimensional, and three-dimensional conduction.
- Plane wall, cylindrical, and spherical conduction.

4.4 Thermal resistance analogy

Conduction can be analysed using a resistance model similar to electrical circuits. This helps simplify composite wall, fin, and insulation problems.

5. Convection

Convection is heat transfer between a solid surface and an adjacent moving fluid, or within a fluid due to both molecular conduction and bulk fluid motion.

Convection can be:

- Natural or free convection, where fluid motion is caused by buoyancy due to density differences.
- Forced convection, where fluid motion is caused by an external device such as a fan, pump, or blower.

5.1 Newton's law of cooling

For convection:

$$q = hA(T_s - T_\infty)$$

where h is the convective heat transfer coefficient, T_s is the surface temperature, and T_∞ is the fluid temperature away from the surface.

5.2 Heat transfer coefficient

The coefficient h depends on fluid properties, flow velocity, geometry, surface roughness, and whether the flow is laminar or turbulent. It is not a material property alone; it is a system property.

5.3 Boundary layer role

In convection, the thermal boundary layer controls the temperature gradient near the wall. A thinner boundary layer usually means stronger heat transfer.

5.4 Engineering examples

- Air cooling of electronic devices.
- Water cooling in radiators.
- Boiling and condensation systems.
- HVAC heating and cooling coils.

6. Radiation

Radiation is heat transfer by electromagnetic waves and does not require a material medium. This is why the Sun can heat the Earth through the vacuum of space.

Radiation depends on temperature and surface properties, especially emissivity. Hotter surfaces radiate more energy, and black or dark surfaces generally emit and absorb more effectively than shiny surfaces.

6.1 Stefan–Boltzmann law

For a blackbody:

$$q = \sigma AT^4$$

For real surfaces, emissivity is included:

$$q = \varepsilon \sigma AT^4$$

where σ is the Stefan–Boltzmann constant.

6.2 Emissivity

Emissivity is the ratio of radiation emitted by a real surface to that emitted by a perfect blackbody at the same temperature. It ranges from 0 to 1.

6.3 Engineering examples

- Solar heat gain.
- Furnace heat transfer.
- Spacecraft thermal control.
- Heat loss from hot pipes and engines.

7. Combined heat transfer

In most real situations, all three mechanisms act together. For example, heat leaving a hot pipe may conduct through the pipe wall, convect into surrounding air, and radiate to nearby objects.

Advanced analysis often requires identifying the dominant mode and then including the others as correction terms or boundary conditions.

8. Heat transfer and thermodynamics

Thermodynamics tells whether heat can flow and how much energy is involved between equilibrium states. Heat transfer determines the time-dependent behavior and the detailed pathway of that energy.

This is why heat transfer is often called a bridge subject between thermodynamics, fluid mechanics, and design.

9. Heat transfer with fluids

When a fluid moves, convection becomes the dominant mode. The motion may be natural, due to buoyancy, or forced, due to pumps and fans.

In many engineering devices, fluid flow and heat transfer are coupled. Faster flow can improve heat transfer, but it can also increase pumping power and pressure drop.

10. Dimensionless numbers

Dimensionless numbers are central to advanced heat transfer because they describe relative effects and help correlate experiments. Important numbers include:

- Reynolds number.
- Prandtl number.
- Nusselt number.
- Grashof number.
- Biot number.
- Fourier number.
- Stefan number.

10.1 Reynolds number

Reynolds number helps determine whether flow is laminar or turbulent, which strongly affects convection.

10.2 Prandtl number

Prandtl number compares momentum diffusivity to thermal diffusivity. It links fluid flow behavior with heat transfer.

10.3 Nusselt number

Nusselt number represents the ratio of convective to conductive heat transfer across a boundary layer. It is widely used in correlation equations.

10.4 Biot number

Biot number compares internal conduction resistance to external convection resistance. It helps decide whether a lumped system analysis is valid.

11. Lumped system analysis

If internal temperature gradients inside a solid are negligible, the object can be treated as a lumped system. This greatly simplifies transient heat transfer problems.

The lumped approach is valid when the Biot number is sufficiently small.

12. Transient heat transfer

Transient heat transfer deals with temperature changing over time. It is important in heating, cooling, quenching, casting, thermal storage, and electronics cooldown.

The analysis often uses time constants, temperature ratios, and dimensionless forms of the heat equation.

13. Heat exchangers

Heat exchangers transfer heat between two fluids without mixing them. They are essential in power plants, chemical plants, engines, refrigerators, and process industries.

13.1 Types

- Parallel flow.
- Counterflow.
- Crossflow.
- Shell-and-tube.
- Plate heat exchangers.

13.2 Performance terms

Important terms include effectiveness, LMTD, overall heat transfer coefficient, and fouling resistance.

14. Extended surfaces

Fins increase surface area and improve heat transfer, especially in convection-dominated systems. They are used in heat sinks, radiators, and engines.

The fin design must balance improved heat transfer against added weight, cost, and material usage.

15. Boiling and condensation

Boiling and condensation are phase-change heat transfer processes that can transfer large amounts of heat. They are very important in refrigeration, air conditioning, steam power plants, and thermal systems.

These processes are often much more efficient than single-phase convection.

16. Thermal design importance

Heat transfer analysis is used to:

- Prevent overheating.
- Improve energy efficiency.
- Size heat exchangers.
- Select insulation.
- Design cooling systems.
- Protect materials from thermal failure.

17. Numerical methods and CFD

Many heat transfer problems do not have simple analytical solutions. Computational methods and CFD are widely used to predict temperature fields, flow behavior, and heat exchanger performance.

18. Typical assumptions

Common simplifying assumptions include:

- Steady or transient behavior.
- One-dimensional heat flow.
- Constant properties.
- Uniform surface temperature.
- Negligible radiation in some cases.

- Perfect contact between surfaces.

These assumptions help with calculation, but they must always be checked against the real system.

19. Common equations summary

Important governing relations are:

- Fourier's law for conduction.
- Newton's law of cooling for convection.
- Stefan–Boltzmann law for radiation.
- Energy balance for transient systems.

20. Applications in mechanical engineering

Heat transfer is used in:

- IC engines.
- Gas turbines.
- Boilers.
- Condensers.
- Refrigerators.
- Air conditioners.
- Heat sinks.
- Furnaces.
- Solar thermal systems.
- Thermal insulation design.

21. Exam-oriented understanding

If you are writing university notes, remember:

- Conduction needs a medium.
- Convection needs fluid motion.
- Radiation does not need a medium.
- Heat flows from higher temperature to lower temperature.
- Real systems often involve all three modes together.

22. Advanced perspective

At advanced level, heat transfer is not just about memorizing formulas. It is about understanding temperature fields, governing equations, boundary conditions, physical interpretation, scaling analysis, and design tradeoffs. Strong engineers use both theory and approximation to solve thermal problems efficiently.

23. Conclusion

Heat transfer is a core mechanical engineering subject that explains how thermal energy moves and how that movement can be controlled in practical systems. Mastery of conduction, convection, radiation, and their combined effects is essential for thermal design, energy systems, and modern engineering analysis.

24. Quick revision points

- Heat transfer is driven by temperature difference.
- Three modes: conduction, convection, radiation.
- Fourier's law governs conduction.
- Newton's law governs convection.
- Stefan–Boltzmann law governs radiation.
- Dimensionless numbers are essential in advanced analysis.
- Heat exchangers and fins are major engineering applications.